VNR Vignana Jyothi Institute of Engineering

Electrical and Electronics Engineering | Course Code: EE201

Electric Circuit Analysis & Network Theorems - Laboratory Manual & Pract

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Syllabus Focus: EE201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Circuit Topologies & Systematic Analysis

" Syllabus Scope: KCL, KVL, graph theory incidence, tie-set, cut-set matrices, nodal and mesh analysis wi

Unit 2: Network Theorems in DC and AC

" Syllabus Scope: Superposition theorem, Thevenin theorem, Norton theorem, Maximum Power Transfer theorem

EE201: Electric Circuit Analysis & Network Theorems - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Transient Response of Dynamic Circuits

" First-order RL and RC circuits, step and impulse response, second-order RLC circuits, overdamped/critically

Unit 4: Sinusoidal Steady-State & Resonant Circuits

" Phasor concepts, impedance and admittance, series and parallel resonance, quality factor Q, bandwidth, 3-pha

Unit 5: Two-Port Network Parameters

" Impedance Z, Admittance Y, Hybrid h, Transmission ABCD parameters, interrelationships, reciprocal and symmet

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Verification of Thevenin, Norton, and Maximum Power Transfer Theorems on DC Test Circuits

Lab 2: Transient Response Measurement of Series RL and RC Circuits using Digital Storage Oscilloscope

Lab 3: Determination of Resonant Frequency, Bandwidth, and Quality Factor of Series and Parallel RLC Circ

Lab 4: Measurement and Verification of Z and Y Parameters for Two-Port Passive Networks

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* William H. Hayt, Jack E. Kemmerly - Engineering Circuit Analysis McGraw-Hill 8th Ed
* Charles K. Alexander, Matthew N.O. Sadiku - Fundamentals of Electric Circuits McGraw-Hill 5th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students